

Unit 1: Lecture Notes

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Course: Calculus and Ordinary Differential Equations

Chapter 1: Differential Calculus (Functions of One Variable)

Course Code: 25MT105

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Detailed Notes: Differentiation of Functions of One Variable

Topic 1: Mean Value Theorems and Applications

Introduction

Mean Value Theorems are fundamental results in calculus that establish relationships between the values of a function and its derivatives. These theorems provide the theoretical foundation for many results in calculus and have important applications in analyzing function behavior.

1.1 Rolle's Theorem

Statement: Let $f(x)$ be a function satisfying:

1. $f(x)$ is continuous on the closed interval $[a, b]$
2. $f(x)$ is differentiable on the open interval (a, b)
3. $f(a) = f(b)$

Then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.

Geometric Interpretation: If a continuous and differentiable curve starts and ends at the same height, there must be at least one point where the tangent is horizontal.

Detailed Example:

Problem: Verify Rolle's theorem for $f(x) = x^2 - 5x + 6$ on $[2, 3]$ and find the value of c .

Solution:

Step 1: Check continuity on $[2, 3]$

Since $f(x) = x^2 - 5x + 6$ is a polynomial, it is continuous everywhere, including on $[2, 3]$. ✓

Step 2: Check differentiability on $(2, 3)$

$$f'(x) = 2x - 5$$

This derivative exists for all x , so $f(x)$ is differentiable on $(2, 3)$. ✓

Step 3: Check if $f(a) = f(b)$

$$f(2) = (2)^2 - 5(2) + 6 = 4 - 10 + 6 = 0$$

$$f(3) = (3)^2 - 5(3) + 6 = 9 - 15 + 6 = 0$$

Since $f(2) = f(3) = 0$, the condition is satisfied. ✓

Step 4: Find c such that $f'(c) = 0$

$$f'(c) = 2c - 5 = 0$$

$$2c = 5$$

$$c = \frac{5}{2} = 2.5$$

Step 5: Verify $c \in (2, 3)$

Since $2 < 2.5 < 3$, we have $c \in (2, 3)$. ✓

Conclusion: All conditions of Rolle's theorem are satisfied, and $c = 2.5$ is the point where the tangent is horizontal.

1.2 Lagrange's Mean Value Theorem (LMVT)

Statement: Let $f(x)$ be a function satisfying:

1. $f(x)$ is continuous on $[a, b]$
2. $f(x)$ is differentiable on (a, b)

Then there exists at least one point $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Geometric Interpretation: The instantaneous rate of change (derivative) at some point equals the average rate of change over the interval.

Note: Rolle's theorem is a special case of LMVT where $f(a) = f(b)$, making the right side zero.

Detailed Example:

Problem: Verify Lagrange's Mean Value Theorem for $f(x) = x^3 - 2x^2 + x + 3$ on $[0, 2]$ and find all values of c .

Solution:

Step 1: Check continuity on $[0, 2]$

$f(x) = x^3 - 2x^2 + x + 3$ is a polynomial, hence continuous on $[0, 2]$. ✓

Step 2: Check differentiability on $(0, 2)$

$$f'(x) = 3x^2 - 4x + 1$$

This exists for all x , so $f(x)$ is differentiable on $(0, 2)$. ✓

Step 3: Calculate $f(a)$ and $f(b)$

$$f(0) = (0)^3 - 2(0)^2 + 0 + 3 = 3$$

$$f(2) = (2)^3 - 2(2)^2 + 2 + 3 = 8 - 8 + 2 + 3 = 5$$

Step 4: Calculate the average rate of change

$$\frac{f(b) - f(a)}{b - a} = \frac{f(2) - f(0)}{2 - 0} = \frac{5 - 3}{2} = \frac{2}{2} = 1$$

Step 5: Find c such that $f'(c) = 1$

$$f'(c) = 3c^2 - 4c + 1 = 1$$

$$3c^2 - 4c + 1 - 1 = 0$$

$$3c^2 - 4c = 0$$

$$c(3c - 4) = 0$$

$$c = 0 \quad \text{or} \quad c = \frac{4}{3}$$

Step 6: Check which values lie in $(0, 2)$

- $c = 0$ is not in the open interval $(0, 2)$ ✗
- $c = \frac{4}{3} \approx 1.333$ is in $(0, 2)$ ✓

Conclusion: Lagrange's MVT is verified, and $c = \frac{4}{3}$ is the point where the instantaneous rate equals the average rate.

1.3 Cauchy's Mean Value Theorem (Generalized MVT)

Statement: Let $f(x)$ and $g(x)$ be two functions satisfying:

1. Both are continuous on $[a, b]$
2. Both are differentiable on (a, b)
3. $g'(x) \neq 0$ for all $x \in (a, b)$

Then there exists at least one point $c \in (a, b)$ such that:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Note: When $g(x) = x$, Cauchy's theorem reduces to Lagrange's MVT.

Detailed Example:

Problem: Verify Cauchy's Mean Value Theorem for $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$ and find the value of c .

Solution:

Step 1: Check continuity on $[1, 2]$

Both $f(x) = x^2$ and $g(x) = x^3$ are polynomials, hence continuous on $[1, 2]$. ✓

Step 2: Check differentiability on $(1, 2)$

$f'(x) = 2x$ — exists for all x ✓

$g'(x) = 3x^2$ — exists for all x ✓

Step 3: Verify $g'(x) \neq 0$ on $(1, 2)$

$g'(x) = 3x^2 \neq 0$ for any $x \in (1, 2)$ since $x > 0$ ✓

Step 4: Calculate function values

$f(1) = (1)^2 = 1, f(2) = (2)^2 = 4$

$g(1) = (1)^3 = 1, g(2) = (2)^3 = 8$

Step 5: Calculate the right-hand side of Cauchy's formula

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f(2) - f(1)}{g(2) - g(1)} = \frac{4 - 1}{8 - 1} = \frac{3}{7}$$

Step 6: Find c such that $\frac{f'(c)}{g'(c)} = \frac{3}{7}$

$$\frac{f'(c)}{g'(c)} = \frac{2c}{3c^2} = \frac{2}{3c}$$

Setting this equal to $\frac{3}{7}$:

$$\frac{2}{3c} = \frac{3}{7}$$

$$2 \times 7 = 3 \times 3c$$

$$14 = 9c$$

$$c = \frac{14}{9} \approx 1.556$$

Step 7: Verify $c \in (1, 2)$

Since $1 < \frac{14}{9} < 2$, we have $c \in (1, 2)$. ✓

Conclusion: Cauchy's Mean Value Theorem is verified with $c = \frac{14}{9}$.

Applications of Mean Value Theorems

Application 1: Proving inequalities

Application 2: Showing that a function is constant if its derivative is zero

Application 3: Approximating function values

Application 4: Proving existence of roots

Topic 2: Indeterminate Forms

Introduction

When evaluating limits, we sometimes encounter expressions whose value cannot be determined directly. These are called **indeterminate forms**. The most common indeterminate forms are:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \times \infty, \quad \infty - \infty, \quad 0^0, \quad 1^\infty, \quad \infty^0$$

L'Hôpital's Rule is the primary tool for evaluating limits of indeterminate forms.

L'Hôpital's Rule

For $\frac{0}{0}$ and $\frac{\infty}{\infty}$ forms:

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists.

Indeterminate Form 1: $\frac{0}{0}$

Detailed Example:

Problem: Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$

Solution:

Step 1: Check if it's an indeterminate form

$$\lim_{x \rightarrow 0} \sin 3x = \sin 0 = 0$$

$$\lim_{x \rightarrow 0} 2x = 0$$

This gives $\frac{0}{0}$, which is indeterminate. ✓

Step 2: Apply L'Hôpital's Rule

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{2x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\sin 3x)}{\frac{d}{dx}(2x)}$$

Step 3: Differentiate numerator and denominator

$$\text{Numerator: } \frac{d}{dx}(\sin 3x) = 3 \cos 3x$$

$$\text{Denominator: } \frac{d}{dx}(2x) = 2$$

Step 4: Evaluate the new limit

$$\lim_{x \rightarrow 0} \frac{3 \cos 3x}{2} = \frac{3 \cos 0}{2} = \frac{3 \times 1}{2} = \frac{3}{2}$$

Answer: $\frac{3}{2}$

Indeterminate Form 2: $\frac{\infty}{\infty}$

Detailed Example:

Problem: Evaluate $\lim_{x \rightarrow \infty} \frac{x^2 + 3x}{2x^2 - 5}$

Solution:

Step 1: Check if it's an indeterminate form

As $x \rightarrow \infty$: numerator $\rightarrow \infty$, denominator $\rightarrow \infty$

This gives $\frac{\infty}{\infty}$, which is indeterminate. ✓

Step 2: Apply L'Hôpital's Rule

$$\lim_{x \rightarrow \infty} \frac{x^2 + 3x}{2x^2 - 5} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(x^2 + 3x)}{\frac{d}{dx}(2x^2 - 5)}$$

Step 3: Differentiate

$$\text{Numerator: } \frac{d}{dx}(x^2 + 3x) = 2x + 3$$

Denominator: $\frac{d}{dx}(2x^2 - 5) = 4x$

Step 4: Evaluate the new limit

$$\lim_{x \rightarrow \infty} \frac{2x + 3}{4x}$$

This is still $\frac{\infty}{\infty}$, so apply L'Hôpital's Rule again:

$$\lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(2x + 3)}{\frac{d}{dx}(4x)} = \lim_{x \rightarrow \infty} \frac{2}{4} = \frac{1}{2}$$

Answer: $\frac{1}{2}$

Indeterminate Form 3: $0 \times \infty$

Strategy: Convert to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form by rewriting as:

$$0 \times \infty = \frac{0}{1/\infty} = \frac{0}{0} \quad \text{or} \quad 0 \times \infty = \frac{\infty}{1/0} = \frac{\infty}{\infty}$$

Detailed Example:

Problem: Evaluate $\lim_{x \rightarrow 0^+} x \ln x$

Solution:

Step 1: Identify the form

As $x \rightarrow 0^+$: $x \rightarrow 0$ and $\ln x \rightarrow -\infty$

This gives $0 \times (-\infty)$, which is indeterminate. ✓

Step 2: Rewrite to get $\frac{\infty}{\infty}$ or $\frac{0}{0}$

$$x \ln x = \frac{\ln x}{1/x}$$

Now as $x \rightarrow 0^+$: $\ln x \rightarrow -\infty$ and $\frac{1}{x} \rightarrow +\infty$

This gives $\frac{-\infty}{\infty}$. ✓

Step 3: Apply L'Hôpital's Rule

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(1/x)}$$

Step 4: Differentiate

Numerator: $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Denominator: $\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = -x^{-2} = -\frac{1}{x^2}$

Step 5: Simplify and evaluate

$$\lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} \frac{1}{x} \times \frac{x^2}{-1} = \lim_{x \rightarrow 0^+} \frac{x^2}{-x} = \lim_{x \rightarrow 0^+} (-x) = 0$$

Answer: 0Indeterminate Form 4: $\infty - \infty$ **Strategy:** Combine terms algebraically to convert to $\frac{0}{0}$ or $\frac{\infty}{\infty}$.**Detailed Example:****Problem:** Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$ **Solution:****Step 1:** Identify the formAs $x \rightarrow 0$: $\frac{1}{x} \rightarrow \infty$ and $\frac{1}{\sin x} \rightarrow \infty$ This gives $\infty - \infty$, which is indeterminate. ✓**Step 2:** Combine into a single fraction

$$\frac{1}{x} - \frac{1}{\sin x} = \frac{\sin x - x}{x \sin x}$$

Step 3: Check the new formAs $x \rightarrow 0$:

- Numerator: $\sin x - x \rightarrow 0 - 0 = 0$
- Denominator: $x \sin x \rightarrow 0 \times 0 = 0$

This gives $\frac{0}{0}$. ✓**Step 4:** Apply L'Hôpital's Rule

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x \sin x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\sin x - x)}{\frac{d}{dx}(x \sin x)}$$

Step 5: DifferentiateNumerator: $\frac{d}{dx}(\sin x - x) = \cos x - 1$ Denominator: $\frac{d}{dx}(x \sin x) = \sin x + x \cos x$ (product rule)**Step 6:** Evaluate

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{\sin x + x \cos x} = \frac{\cos 0 - 1}{\sin 0 + 0 \times \cos 0} = \frac{1 - 1}{0 + 0} = \frac{0}{0}$$

Still indeterminate! Apply L'Hôpital's Rule again:

Step 7: Differentiate again

$$\text{Numerator: } \frac{d}{dx}(\cos x - 1) = -\sin x$$

$$\text{Denominator: } \frac{d}{dx}(\sin x + x \cos x) = \cos x + \cos x - x \sin x = 2 \cos x - x \sin x$$

Step 8: Evaluate

$$\lim_{x \rightarrow 0} \frac{-\sin x}{2 \cos x - x \sin x} = \frac{-\sin 0}{2 \cos 0 - 0 \times \sin 0} = \frac{0}{2 \times 1 - 0} = \frac{0}{2} = 0$$

Answer: 0Indeterminate Form 5: 1^∞ **Strategy:** Use the formula: If $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = \infty$, then:

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} g(x)[f(x)-1]}$$

Or use: $y = [f(x)]^{g(x)} \Rightarrow \ln y = g(x) \ln f(x)$ **Detailed Example:****Problem:** Evaluate $\lim_{x \rightarrow 0} (1+x)^{1/x}$ **Solution:****Step 1:** Identify the form

$$\text{As } x \rightarrow 0: (1+x) \rightarrow 1 \text{ and } \frac{1}{x} \rightarrow \infty$$

This gives 1^∞ , which is indeterminate. ✓**Step 2:** Take the natural logarithm

$$\text{Let } y = (1+x)^{1/x}$$

$$\ln y = \ln[(1+x)^{1/x}] = \frac{1}{x} \ln(1+x) = \frac{\ln(1+x)}{x}$$

Step 3: Find $\lim_{x \rightarrow 0} \ln y$

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$$

$$\text{As } x \rightarrow 0: \ln(1+x) \rightarrow \ln 1 = 0 \text{ and } x \rightarrow 0$$

This gives $\frac{0}{0}$. ✓**Step 4:** Apply L'Hôpital's Rule

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} [\ln(1+x)]}{\frac{d}{dx} (x)}$$

Numerator: $\frac{d}{dx}[\ln(1+x)] = \frac{1}{1+x}$

Denominator: $\frac{d}{dx}(x) = 1$

Step 5: Evaluate

$$\lim_{x \rightarrow 0} \frac{1/(1+x)}{1} = \lim_{x \rightarrow 0} \frac{1}{1+x} = \frac{1}{1+0} = 1$$

So $\lim_{x \rightarrow 0} \ln y = 1$

Step 6: Find the original limit

Since $\ln y \rightarrow 1$, we have:

$$y = e^{\ln y} \rightarrow e^1 = e$$

Answer: e

Indeterminate Form 6: 0^0

Strategy: Similar to 1^∞ , use logarithms: $y = [f(x)]^{g(x)} \Rightarrow \ln y = g(x) \ln f(x)$

Detailed Example:

Problem: Evaluate $\lim_{x \rightarrow 0^+} x^x$

Solution:

Step 1: Identify the form

As $x \rightarrow 0^+$: base $x \rightarrow 0$ and exponent $x \rightarrow 0$

This gives 0^0 , which is indeterminate. ✓

Step 2: Take logarithm

Let $y = x^x$

$$\ln y = \ln(x^x) = x \ln x$$

Step 3: Find $\lim_{x \rightarrow 0^+} \ln y$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x$$

This is the $0 \times \infty$ form (we solved this earlier!)

From our previous work: $\lim_{x \rightarrow 0^+} x \ln x = 0$

Step 4: Find the original limit

Since $\ln y \rightarrow 0$:

$$y = e^{\ln y} \rightarrow e^0 = 1$$

Answer: 1

Indeterminate Form 7: ∞^0

Strategy: Use logarithms as before.

Detailed Example:

Problem: Evaluate $\lim_{x \rightarrow \infty} x^{1/x}$

Solution:

Step 1: Identify the form

As $x \rightarrow \infty$: base $x \rightarrow \infty$ and exponent $\frac{1}{x} \rightarrow 0$

This gives ∞^0 , which is indeterminate. ✓

Step 2: Take logarithm

Let $y = x^{1/x}$

$$\ln y = \ln(x^{1/x}) = \frac{1}{x} \ln x = \frac{\ln x}{x}$$

Step 3: Find $\lim_{x \rightarrow \infty} \ln y$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

As $x \rightarrow \infty$: $\ln x \rightarrow \infty$ and $x \rightarrow \infty$

This gives $\frac{\infty}{\infty}$. ✓

Step 4: Apply L'Hôpital's Rule

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x)} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Step 5: Find the original limit

Since $\ln y \rightarrow 0$:

$$y = e^{\ln y} \rightarrow e^0 = 1$$

Answer: 1

Topic 3: Taylor's and Maclaurin's Series Expansion with Remainders

Introduction

Taylor series allow us to represent functions as infinite polynomials, which is extremely useful for approximations, solving differential equations, and analyzing function behavior.

3.1 Taylor Series

Definition: If a function $f(x)$ has derivatives of all orders at $x = a$, then the **Taylor series** of $f(x)$ about $x = a$ is:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

General Form:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

where $f^{(n)}(a)$ denotes the n -th derivative of f evaluated at $x = a$.

3.2 Maclaurin Series

Definition: A **Maclaurin series** is a special case of Taylor series where $a = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

General Form:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

3.3 Common Maclaurin Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}, \quad |x| < 1$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

3.4 Taylor's Theorem with Remainder

When we approximate a function using only the first n terms of its Taylor series, there is an error. The remainder term quantifies this error.

Taylor's Formula with Lagrange Remainder:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x)$$

where the **Lagrange form of the remainder** is:

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}$$

for some c between a and x .

For Maclaurin series ($a = 0$):

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1}$$

for some c between 0 and x .

Detailed Example: Maclaurin Series

Problem: Find the Maclaurin series for $f(x) = e^{2x}$ up to the term containing x^4 , and use it to approximate $e^{0.2}$. Estimate the error using the remainder term.

Solution:

Part 1: Find the Maclaurin series

Step 1: Calculate derivatives and evaluate at $x = 0$

$$f(x) = e^{2x} \quad f(0) = e^0 = 1$$

$$f'(x) = 2e^{2x} \quad f'(0) = 2e^0 = 2$$

$$f''(x) = 4e^{2x} \quad f''(0) = 4e^0 = 4$$

$$f'''(x) = 8e^{2x} \quad f'''(0) = 8e^0 = 8$$

$$f^{(4)}(x) = 16e^{2x} \quad f^{(4)}(0) = 16e^0 = 16$$

$$f^{(5)}(x) = 32e^{2x} \quad f^{(5)}(0) = 32e^0 = 32$$

Step 2: Write the Maclaurin series up to x^4

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \dots$$

$$e^{2x} = 1 + 2x + \frac{4}{2}x^2 + \frac{8}{6}x^3 + \frac{16}{24}x^4 + \dots$$

$$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \dots$$

Part 2: Approximate $e^{0.2}$

We want to approximate $e^{2(0.1)} = e^{0.2}$ using $x = 0.1$:

$$\begin{aligned}
 e^{0.2} &\approx 1 + 2(0.1) + 2(0.1)^2 + \frac{4}{3}(0.1)^3 + \frac{2}{3}(0.1)^4 \\
 &= 1 + 0.2 + 2(0.01) + \frac{4}{3}(0.001) + \frac{2}{3}(0.0001) \\
 &= 1 + 0.2 + 0.02 + \frac{0.004}{3} + \frac{0.0002}{3} \\
 &= 1 + 0.2 + 0.02 + 0.001333... + 0.0000667... \\
 &\approx 1.2214
 \end{aligned}$$

Part 3: Estimate the error

The error is given by the remainder term:

$$R_4(x) = \frac{f^{(5)}(c)}{5!}x^5$$

where c is between 0 and $x = 0.1$.

$$R_4(0.1) = \frac{32e^{2c}}{120}(0.1)^5 = \frac{32e^{2c}}{120}(0.00001)$$

Since $0 < c < 0.1$, we have $1 < e^{2c} < e^{0.2} < 1.25$ (approximately).

Maximum error:

$$|R_4(0.1)| < \frac{32 \times 1.25}{120} \times 0.00001 = \frac{40}{120} \times 0.00001 = \frac{1}{3} \times 0.00001 \approx 0.0000033$$

Answer:

- Maclaurin series: $e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \dots$
- Approximation: $e^{0.2} \approx 1.2214$
- Maximum error: < 0.0000034

(Actual value: $e^{0.2} \approx 1.2214027...$, so our approximation is very accurate!)

Topic 4: Concavity and Convexity of a Curve

Definitions

Concave Up (Convex): A function $f(x)$ is **concave up** (or convex) on an interval if its graph lies above all its tangent lines on that interval. Geometrically, the curve "holds water."

Concave Down (Concave): A function $f(x)$ is **concave down** (or concave) on an interval if its graph lies below all its tangent lines on that interval. Geometrically, the curve "spills water."

Mathematical Test for Concavity

Second Derivative Test:

1. If $f''(x) > 0$ on an interval, then $f(x)$ is **concave up** on that interval.
 2. If $f''(x) < 0$ on an interval, then $f(x)$ is **concave down** on that interval.
-

Detailed Example 1: Concavity

Problem: Determine the intervals where $f(x) = x^3 - 6x^2 + 9x + 1$ is concave up and concave down.

Solution:

Step 1: Find the first derivative

$$f'(x) = 3x^2 - 12x + 9$$

Step 2: Find the second derivative

$$f''(x) = 6x - 12$$

Step 3: Find where $f''(x) = 0$

$$6x - 12 = 0$$

$$6x = 12$$

$$x = 2$$

Step 4: Test the sign of $f''(x)$ on intervals divided by $x = 2$

Interval 1: $x < 2$ (test point: $x = 0$)

$$f''(0) = 6(0) - 12 = -12 < 0$$

So f is **concave down** on $(-\infty, 2)$.

Interval 2: $x > 2$ (test point: $x = 3$)

$$f''(3) = 6(3) - 12 = 18 - 12 = 6 > 0$$

So f is **concave up** on $(2, \infty)$.

Answer:

- **Concave up:** $(2, \infty)$
 - **Concave down:** $(-\infty, 2)$
-

Detailed Example 2: Concavity

Problem: Determine the concavity of $f(x) = e^{-x^2}$.

Solution:

Step 1: Find the first derivative

$$f'(x) = e^{-x^2} \cdot (-2x) = -2xe^{-x^2}$$

Step 2: Find the second derivative (using product rule)

$$\begin{aligned}
 f''(x) &= \frac{d}{dx}[-2xe^{-x^2}] \\
 &= -2e^{-x^2} + (-2x)(-2x)e^{-x^2} \\
 &= -2e^{-x^2} + 4x^2e^{-x^2} \\
 &= e^{-x^2}(4x^2 - 2) \\
 &= 2e^{-x^2}(2x^2 - 1)
 \end{aligned}$$

Step 3: Find where $f''(x) = 0$

Since $e^{-x^2} > 0$ for all x :

$$\begin{aligned}
 2x^2 - 1 &= 0 \\
 x^2 &= \frac{1}{2} \\
 x &= \pm \frac{1}{\sqrt{2}} = \pm \frac{\sqrt{2}}{2}
 \end{aligned}$$

Step 4: Test the sign of $f''(x)$

Interval 1: $x < -\frac{\sqrt{2}}{2}$ (test: $x = -1$)

$$f''(-1) = 2e^{-1}(2(1) - 1) = 2e^{-1}(1) > 0$$

Concave up

Interval 2: $-\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$ (test: $x = 0$)

$$f''(0) = 2e^0(2(0) - 1) = 2(1)(-1) = -2 < 0$$

Concave down

Interval 3: $x > \frac{\sqrt{2}}{2}$ (test: $x = 1$)

$$f''(1) = 2e^{-1}(2(1) - 1) = 2e^{-1}(1) > 0$$

Concave up

Answer:

- **Concave up:** $\left(-\infty, -\frac{\sqrt{2}}{2}\right) \cup \left(\frac{\sqrt{2}}{2}, \infty\right)$
- **Concave down:** $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

Topic 5: Points of Inflection

Definition

A **point of inflection** is a point on the curve where the concavity changes (from concave up to concave down, or vice versa).

Finding Points of Inflection

Steps:

1. Find $f''(x)$
2. Solve $f''(x) = 0$ or find where $f''(x)$ is undefined
3. Check if the concavity actually changes at these points
4. If concavity changes, $(c, f(c))$ is a point of inflection

Note: $f''(c) = 0$ is necessary but not sufficient. The second derivative must change sign.

Detailed Example 1: Points of Inflection

Problem: Find all points of inflection for $f(x) = x^4 - 4x^3$.

Solution:

Step 1: Find the second derivative

$$f'(x) = 4x^3 - 12x^2$$

$$f''(x) = 12x^2 - 24x = 12x(x - 2)$$

Step 2: Solve $f''(x) = 0$

$$12x(x - 2) = 0$$

$$x = 0 \quad \text{or} \quad x = 2$$

Step 3: Check if concavity changes at these points

At $x = 0$:

Test $x = -1$: $f''(-1) = 12(-1)(-1 - 2) = 12(-1)(-3) = 36 > 0$ (concave up)

Test $x = 1$: $f''(1) = 12(1)(1 - 2) = 12(1)(-1) = -12 < 0$ (concave down)

Concavity changes from up to down ✓ **Point of inflection**

At $x = 2$:

Test $x = 1$: $f''(1) = -12 < 0$ (concave down)

Test $x = 3$: $f''(3) = 12(3)(3 - 2) = 12(3)(1) = 36 > 0$ (concave up)

Concavity changes from down to up ✓ **Point of inflection**

Step 4: Find the y-coordinates

$$f(0) = 0^4 - 4(0)^3 = 0$$

$$f(2) = 2^4 - 4(2)^3 = 16 - 32 = -16$$

Answer: Points of inflection are $(0, 0)$ and $(2, -16)$.

Detailed Example 2: Points of Inflection

Problem: Find all points of inflection for $f(x) = \sin x$ on $[0, 2\pi]$.

Solution:

Step 1: Find the second derivative

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

Step 2: Solve $f''(x) = 0$

$$-\sin x = 0$$

$$\sin x = 0$$

On $[0, 2\pi]$: $x = 0, \pi, 2\pi$

Step 3: Check if concavity changes

At $x = 0$:

Test $x = -0.1$ (slightly left): $f''(-0.1) = -\sin(-0.1) \approx 0.0998 > 0$

Test $x = 0.1$ (slightly right): $f''(0.1) = -\sin(0.1) \approx -0.0998 < 0$

Concavity changes ✓ **Point of inflection at $x = 0$** (boundary point)

At $x = \pi$:

Test $x = 3$ (slightly left, since $\pi \approx 3.14$): $f''(3) = -\sin(3) \approx -0.141 < 0$

Test $x = 3.5$ (slightly right): $f''(3.5) = -\sin(3.5) \approx 0.351 > 0$

Concavity changes ✓ **Point of inflection**

At $x = 2\pi$:

Test $x = 6$ (slightly left, since $2\pi \approx 6.28$): $f''(6) = -\sin(6) \approx 0.279 > 0$

Test $x = 6.5$ (slightly right): $f''(6.5) = -\sin(6.5) \approx -0.215 < 0$

Concavity changes ✓ **Point of inflection at $x = 2\pi$** (boundary point)

Step 4: Find y-coordinates

$$f(0) = \sin(0) = 0$$

$$f(\pi) = \sin(\pi) = 0$$

$$f(2\pi) = \sin(2\pi) = 0$$

Answer: Points of inflection are $(0, 0)$, $(\pi, 0)$, and $(2\pi, 0)$.

Topic 6: Curve Sketching

Introduction

Curve sketching involves analyzing a function systematically to draw its graph accurately. We use derivatives and other tools to determine key features.

Steps for Curve Sketching

Step 1: Domain

- Find the domain of $f(x)$ (where is it defined?)

Step 2: Intercepts

- **y-intercept:** Set $x = 0$, find $f(0)$
- **x-intercepts:** Set $f(x) = 0$, solve for x

Step 3: Symmetry

- **Even function:** $f(-x) = f(x)$ (symmetric about y-axis)
- **Odd function:** $f(-x) = -f(x)$ (symmetric about origin)

Step 4: Asymptotes

- **Vertical asymptotes:** Where denominator = 0 (for rational functions)
- **Horizontal asymptotes:** $\lim_{x \rightarrow \pm\infty} f(x)$
- **Oblique asymptotes:** If degree of numerator = degree of denominator + 1

Step 5: First Derivative Analysis

- Find $f'(x)$
- Find critical points: Solve $f'(x) = 0$ or where $f'(x)$ is undefined
- Determine intervals of increase/decrease
- Identify local maxima and minima

Step 6: Second Derivative Analysis

- Find $f''(x)$
- Determine concavity: Where is $f''(x) > 0$ or $f''(x) < 0$?
- Find inflection points: Where does $f''(x) = 0$ and concavity changes?

Step 7: Sketch the Curve

- Plot all key points (intercepts, extrema, inflection points)
 - Draw asymptotes
 - Connect points smoothly according to increase/decrease and concavity
-

Detailed Example 1: Complete Curve Sketching

Problem: Sketch the curve $f(x) = \frac{x^2}{x-1}$.

Solution:

Step 1: Domain

The function is undefined when $x - 1 = 0$, i.e., $x = 1$.

Domain: $(-\infty, 1) \cup (1, \infty)$

Step 2: Intercepts

y-intercept: $f(0) = \frac{0^2}{0-1} = \frac{0}{-1} = 0$ Point: $(0, 0)$

x-intercepts: Set $f(x) = 0$

$$\frac{x^2}{x-1} = 0 \Rightarrow x^2 = 0 \Rightarrow x = 0$$

Point: $(0, 0)$

Step 3: Symmetry

$$f(-x) = \frac{(-x)^2}{-x-1} = \frac{x^2}{-x-1}$$

This is neither $f(x)$ nor $-f(x)$, so **no symmetry**.

Step 4: Asymptotes

Vertical asymptote: At $x = 1$ (where denominator = 0)

Check behavior near $x = 1$:

- As $x \rightarrow 1^-$: numerator $\rightarrow 1$, denominator $\rightarrow 0^-$, so $f(x) \rightarrow -\infty$
- As $x \rightarrow 1^+$: numerator $\rightarrow 1$, denominator $\rightarrow 0^+$, so $f(x) \rightarrow +\infty$

Horizontal asymptote:

$$\lim_{x \rightarrow \infty} \frac{x^2}{x-1}$$

Degree of numerator (2) > degree of denominator (1), so **no horizontal asymptote**.

Oblique asymptote:

Perform polynomial long division:

$$\frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}$$

As $x \rightarrow \pm\infty$, $\frac{1}{x-1} \rightarrow 0$, so the oblique asymptote is $y = x + 1$.

Step 5: First Derivative Analysis

Using quotient rule:

$$f'(x) = \frac{(x-1)(2x) - x^2(1)}{(x-1)^2} = \frac{2x^2 - 2x - x^2}{(x-1)^2} = \frac{x^2 - 2x}{(x-1)^2}$$

$$f'(x) = \frac{x(x-2)}{(x-1)^2}$$

Critical points: $f'(x) = 0$

$$x(x-2) = 0 \Rightarrow x = 0 \text{ or } x = 2$$

Sign analysis of $f'(x)$:

Interval	$(-\infty, 0)$	$(0, 1)$	$(1, 2)$	$(2, \infty)$
x	—	+	+	+
$x - 2$	—	—	—	+
$(x - 1)^2$	+	+	+	+
$f'(x)$	+	—	—	+
$f(x)$	Increasing	Decreasing	Decreasing	Increasing

Local maximum at $x = 0$: $f(0) = 0$, point $(0, 0)$

Local minimum at $x = 2$: $f(2) = \frac{4}{1} = 4$, point $(2, 4)$

Step 6: Second Derivative Analysis

This gets complex. Let's use the quotient rule on $f'(x) = \frac{x^2-2x}{(x-1)^2}$:

$$f''(x) = \frac{(x-1)^2(2x-2) - (x^2-2x) \cdot 2(x-1)}{(x-1)^4}$$

$$= \frac{(x-1)[(x-1)(2x-2) - 2(x^2-2x)]}{(x-1)^4}$$

$$= \frac{(x-1)(2x-2) - 2(x^2-2x)}{(x-1)^3}$$

$$= \frac{2x^2 - 2x - 2x + 2 - 2x^2 + 4x}{(x-1)^3}$$

$$= \frac{2}{(x-1)^3}$$

Concavity:

$$f''(x) = \frac{2}{(x-1)^3}$$

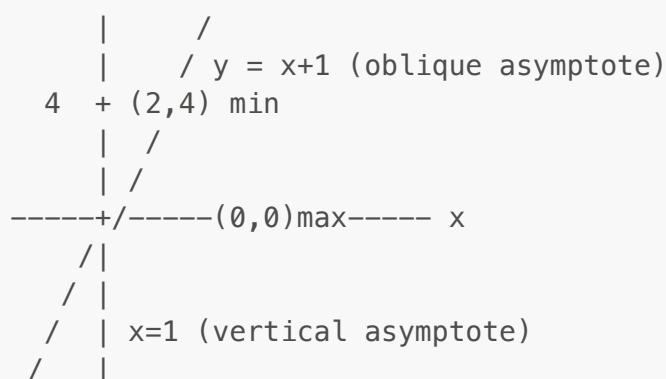
- For $x < 1$: $(x-1)^3 < 0$, so $f''(x) < 0 \rightarrow$ **concave down**
- For $x > 1$: $(x-1)^3 > 0$, so $f''(x) > 0 \rightarrow$ **concave up**

Inflection points: $f''(x)$ is never zero, only undefined at $x = 1$ (which is not in domain).

No inflection points.

Step 7: Sketch**Summary of key features:**

- Domain: $(-\infty, 1) \cup (1, \infty)$
- Intercepts: $(0, 0)$
- Vertical asymptote: $x = 1$
- Oblique asymptote: $y = x + 1$
- Local max: $(0, 0)$
- Local min: $(2, 4)$
- Increasing on: $(-\infty, 0) \cup (2, \infty)$
- Decreasing on: $(0, 1) \cup (1, 2)$
- Concave down: $(-\infty, 1)$
- Concave up: $(1, \infty)$

Sketch:**Detailed Example 2: Complete Curve Sketching**

Problem: Sketch the curve $f(x) = x^3 - 3x^2 + 2$.

Solution:**Step 1: Domain**

Polynomial function, defined everywhere.

Domain: $(-\infty, \infty)$

Step 2: Intercepts

y-intercept: $f(0) = 0 - 0 + 2 = 2$ Point: $(0, 2)$

x-intercepts: Set $f(x) = 0$

$$x^3 - 3x^2 + 2 = 0$$

Try $x = 1$: $1 - 3 + 2 = 0$ ✓

Factor: $(x - 1)(x^2 - 2x - 2) = 0$

Using quadratic formula on $x^2 - 2x - 2 = 0$:

$$x = \frac{2 \pm \sqrt{4 + 8}}{2} = \frac{2 \pm \sqrt{12}}{2} = \frac{2 \pm 2\sqrt{3}}{2} = 1 \pm \sqrt{3}$$

x-intercepts: $x = 1$, $x = 1 + \sqrt{3} \approx 2.73$, $x = 1 - \sqrt{3} \approx -0.73$

Step 3: Symmetry

$$f(-x) = (-x)^3 - 3(-x)^2 + 2 = -x^3 - 3x^2 + 2$$

Not equal to $f(x)$ or $-f(x)$, so **no symmetry**.

Step 4: Asymptotes

Polynomial $\rightarrow$ **no asymptotes**

End behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Step 5: First Derivative Analysis

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

Critical points: $f'(x) = 0$

$$3x(x - 2) = 0 \Rightarrow x = 0 \text{ or } x = 2$$

Sign analysis:

Interval	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
$f'(x)$	+	-	+
$f(x)$	Increasing	Decreasing	Increasing

Local maximum at $x = 0$: $f(0) = 2$, point $(0, 2)$

Local minimum at $x = 2$: $f(2) = 8 - 12 + 2 = -2$, point $(2, -2)$

Step 6: Second Derivative Analysis

$$f''(x) = 6x - 6 = 6(x - 1)$$

Find where $f''(x) = 0$:

$$6(x - 1) = 0 \Rightarrow x = 1$$

Concavity:

Interval	$(-\infty, 1)$	$(1, \infty)$
$f''(x)$	-	+
Concavity	Down	Up

Inflection point at $x = 1$:

$$f(1) = 1 - 3 + 2 = 0$$

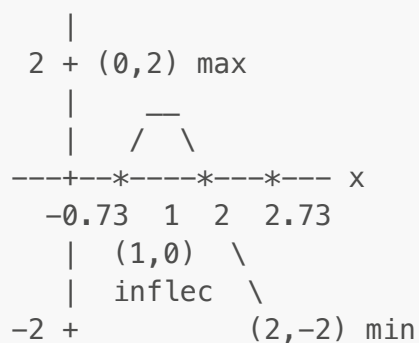
Point: $(1, 0)$

Step 7: Sketch

Summary:

- Domain: All real numbers
- y-intercept: $(0, 2)$
- x-intercepts: $(-0.73, 0)$, $(1, 0)$, $(2.73, 0)$
- Local max: $(0, 2)$
- Local min: $(2, -2)$
- Inflection point: $(1, 0)$
- Increasing: $(-\infty, 0) \cup (2, \infty)$
- Decreasing: $(0, 2)$
- Concave down: $(-\infty, 1)$
- Concave up: $(1, \infty)$

Sketch:



The curve starts from bottom left, crosses x-axis at -0.73 , reaches maximum at $(0, 2)$, passes through inflection point at $(1, 0)$ where concavity changes, reaches minimum at $(2, -2)$, crosses x-axis at 2.73 , and continues to top right.

Summary

These six topics form the foundation of analyzing and understanding the behavior of functions using calculus:

1. **Mean Value Theorems** establish fundamental relationships between function values and derivatives
2. **Indeterminate Forms** provide techniques for evaluating challenging limits
3. **Taylor/Maclaurin Series** allow polynomial approximations of functions
4. **Concavity** describes how curves bend
5. **Inflection Points** mark where curves change their bending direction
6. **Curve Sketching** synthesizes all these tools to visualize functions completely

These concepts are essential for advanced calculus, optimization, numerical methods, and applications in engineering and physics.